

A white Emirates airplane is flying diagonally across a clear blue sky, leaving four thick, parallel white smoke trails behind it. The aircraft is viewed from below, showing its four engines and the 'Emirates' logo on the tail.

ME 300:
Engineering Thermodynamics

Steady Flow Energy Equation

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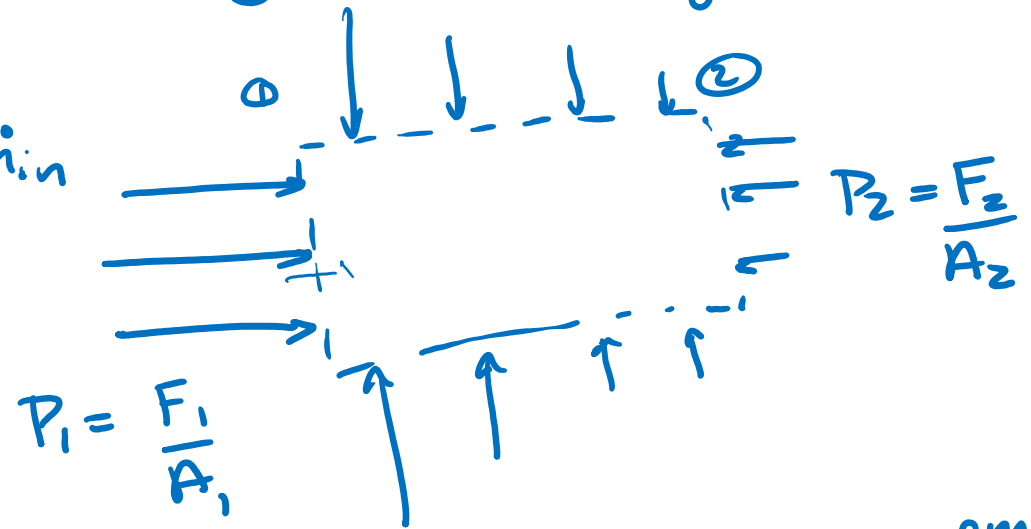
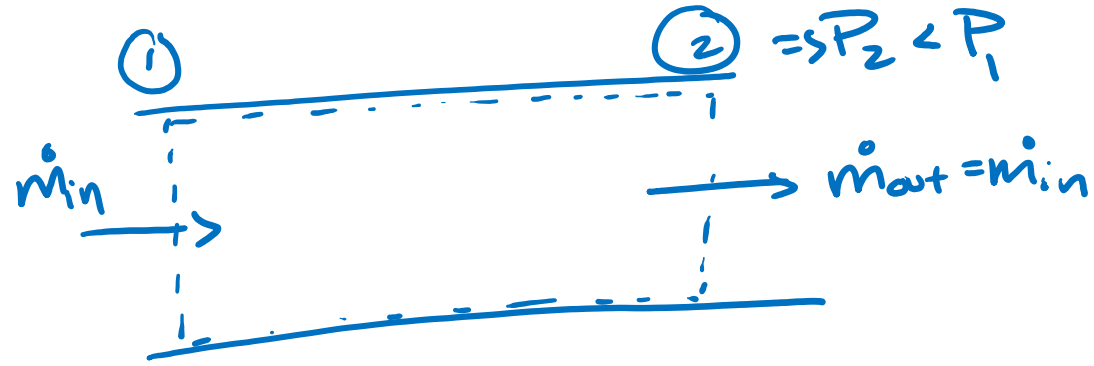
↳ Instantaneous: $\frac{dE_{cv}}{dt} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_{gen}$ $\sim [W], [J/s]$

$\Rightarrow$ ★ Steady flow assumption: $\frac{d}{dt} = 0 \Rightarrow \boxed{\dot{E}_{in} = \dot{E}_{out}}$
 $E_{cv} = \text{const}$

$\Rightarrow$ change: heat, work
 $\dot{Q}, \dot{W}$

$\Rightarrow \dot{m}_{in} = \dot{m}_{out}$
 $\dot{m}_{cv} = \text{const.}$

Flow work $\rightarrow$ work associated with moving fluid through a C.V.



$$\rightarrow \dot{W} = \frac{\text{work}}{\text{time}} = \frac{\text{force} \times \text{distance}}{\text{time}}$$

$$\Rightarrow \dot{W}_{\text{flow}} = P A v = \frac{P}{\rho} A \rho v P \rightarrow \boxed{\dot{W}_{\text{flow}} = \dot{m} P v}$$

$$\Rightarrow \dot{W}_{\text{flow},1} = \dot{m} P_1 v_1, \quad \dot{W}_{\text{flow},2} = -\dot{m} P_2 v_2$$

$[m^3/kg]$

Steady Flow Energy Equation $e = \frac{E}{M} = ke + pe + u \sim [J/kg]$

$$\rightarrow \dot{E}_{in} = \dot{E}_{out}$$

$$\dot{Q}_{in} + \dot{W}_{in, \text{ext}} + \dot{m}e_{in} = \dot{Q}_{out} + \dot{W}_{out, \text{ext}} + \dot{m}e_{out}$$

$$\dot{Q}_{in} + \dot{W}_{flow, in} + \dot{W}_{in} + \dot{m}e_{in} = \dot{Q}_{out} + \dot{W}_{flow, out} + \dot{W}_{out} + \dot{m}e_{out}$$

$$\Rightarrow \dot{m}(e_{out} - e_{in}) = \dot{Q}_{in, net} - \dot{W}_{out, net} + \underbrace{\dot{W}_{flow}}_{\dot{m}(P_1 v_1 - P_2 v_2)}$$

$$h = u + Pv$$

$$\dot{m}(e_2 + P_2 v_2 - e_1 - P_1 v_1) = \dot{Q}_{in, net} - \dot{W}_{out, net}$$

$$\dot{m}(u_2 + ke_2 + pe_2 + P_2 v_2) - (u_1 + ke_1 + pe_1 + P_1 v_1) = \dot{Q}_{in, net} - \dot{W}_{out, net}$$

$$\boxed{\dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}}$$



For a given mass flow, does the enthalpy of a fluid increase, decrease, or stay the same if heat is added?

$$\dot{m}(\underbrace{\Delta h}_{\uparrow} + \Delta pe) = \underbrace{\dot{Q}}_{\uparrow} - \dot{W}$$

- A. Increase
- B. Decrease
- C. Stay the same

Example: Nozzle flow

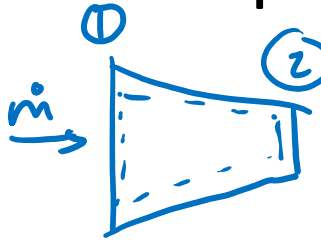


What is the point of a nozzle?

A. Increase velocity

B. Decrease velocity

Example: Nozzle flow



② $A_2 < A_1$

$$T_1 = 300 \text{ K}$$

$$v_1 = 10 \text{ m/s}$$

$$c_p = 1001 \text{ J/kg}\cdot\text{K}$$

$$v_2 = 250 \text{ m/s}$$

$$T_2 = ?$$

Assume

- steady-flow
- $\Delta p_e = 0$, $\dot{W} = 0$, $\dot{Q} = 0$
- air = ideal gas
 $\hookrightarrow c_p = \text{const}$

$$\dot{m} (\Delta h + \Delta ke + \cancel{\Delta pe}) = \cancel{\dot{Q}} - \cancel{\dot{W}} \rightarrow$$

$$\downarrow \quad \uparrow$$
$$\Delta h + \Delta ke = 0$$

$$c_p \Delta T + \Delta ke = 0$$

$$T_2 = T_1 - \frac{1}{2c_p} (v_2^2 - v_1^2)$$

$$= 300 - \frac{1}{2(1001)} (250^2 - 10^2)$$

$$\boxed{T_2 = 269 \text{ K}}$$